

Lab #2: DMM Ohmmeter Measurement and the Breadboard

Pre-Lab Questions

Q1 What is the purpose of a Digital Multimeter?

Q2 When measuring resistance using a DMM, should the component or circuit be powered or unpowered? Briefly state your reason.

Q3 Which resistance-measurement mode, 2W (two-wire) or 4W (four-wire), is generally more accurate when measuring very small resistances? Briefly explain.

Q4 In a 2W (two-wire) resistance measurement, if the two DMM test leads are clipped together, would you expect the measured resistance to be exactly $0\ \Omega$? Briefly explain.

Q5 A resistor is labeled $1\ \text{k}\Omega$ with a tolerance of 5%. What is the minimum possible resistance?

Q6 A resistor is labeled $1\ \text{k}\Omega$ with a tolerance of 5%. What is the maximum possible resistance?

Q7 What does the continuity function of a DMM test?

Q8 In continuity mode, what does it usually mean when the DMM beeps?